

Quadratics: one curve, three forms, and VAR

THE ONE IDEA

A quadratic is an equation with one variable squared. It is traditionally written in three forms, and each form gives you one fact for free.

EXAMPLE

$$y = \underbrace{2x^2 - 12x + 10}_{\text{standard form}} = \underbrace{2(x-1)(x-5)}_{\text{factorised form}} = \underbrace{2(x-3)^2 - 8}_{\text{vertex form}}$$

THE THREE FORMS IN GENERAL, AND THE FACT EACH ONE HANDS YOU

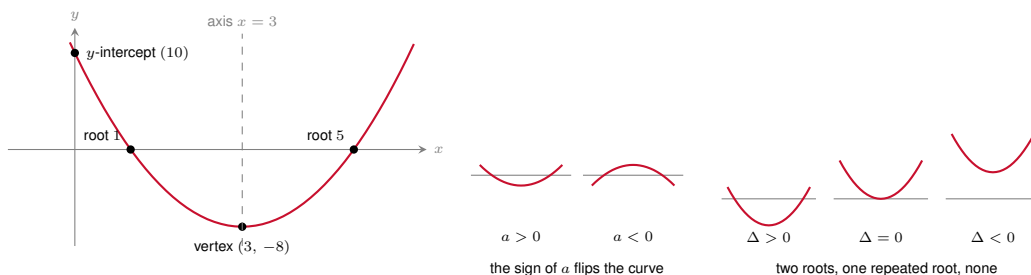
Form	Formula	What it hands you	In the example
Standard	$y = ax^2 + bx + c$	Crosses the y -axis at c . The sign of a says which way it opens.	$2x^2 - 12x + 10$: y -intercept 10; $a = 2 > 0$, opens upward.
Factorised	$y = a(x-r_1)(x-r_2)$	Crosses the x -axis at the roots r_1, r_2 . The vertex sits halfway between them.	$2(x-1)(x-5)$: roots 1 and 5; axis of symmetry $x = 3$.
Vertex	$y = a(x-h)^2 + k$	Vertex at (h, k) : a minimum if $a > 0$ (concave up), a maximum if $a < 0$ (concave down).	$2(x-3)^2 - 8$: minimum value -8 , at $x = 3$.

WHERE THE DISCRIMINANT COMES FROM

$$x = \frac{-b \pm \sqrt{\underbrace{b^2 - 4ac}_{\Delta}}}{2a}$$

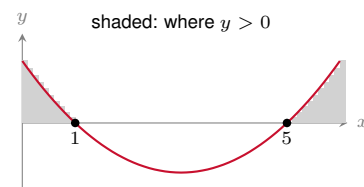
The roots come from the quadratic formula. The part under the square root is called the discriminant, $\Delta = b^2 - 4ac$. Its sign decides how many real roots there are: $\Delta > 0$ two, $\Delta = 0$ one repeated (the curve touches), $\Delta < 0$ none. In the example $\Delta = 144 - 80 = 64 > 0$.

THE EXAMPLE, THE SIGN OF A, AND THE THREE CASES OF THE DISCRIMINANT



INEQUALITIES

$2x^2 - 12x + 10 > 0$ where the curve is above the axis: outside the roots, $x < 1$ or $x > 5$, written $]-\infty, 1[\cup]5, +\infty[$. For ≤ 0 : between them, $x \in [1, 5]$. Read the sign off the picture.



IN PLAIN ENGLISH

The factorised form's free gift: the top of the flight is halfway between kick and landing. $x = -b/(2a)$ gives the same line, and still works with no roots to average.

TRAPS

- The minus sign in $x = -b/(2a)$. Drop it and the vertex lands on the wrong side.
- " > 0 " versus " ≥ 0 ". The equals sign allows $\Delta = 0$, one touching root.
- " $f(x) > 0$ for all x " needs two things: $\Delta < 0$ and $a > 0$. In the notes' example $a = 2$ is given, so only Δ matters.

LISTEN FOR THESE THREE MOMENTS

1. The discriminant cases are sorted on the board. Draw the three pictures beside the three signs.
2. The vertex is found: $-b/(2a)$ or the midpoint of the roots. They agree.
3. The vertex form arrives. He shows both ways. Notice which he says to prefer.

Questions to ask in class

Ask at least one. The first is safe, the second shows you followed, the third is the one you actually want answered. Write down what was said; it is the part you will remember.

1. *[safe]*

If $\Delta < 0$ there are no roots to take the midpoint of. Is the vertex still at $x = -b/(2a)$?

What you expect: Yes. The symmetry does not depend on the roots existing. It gets the symmetry said out loud, which is the idea the whole section rests on.

What he said:

2. *[shows you followed]*

In the example where we find k , $a = 2$ is fixed. If a were unknown as well, would " $f(x) > 0$ for all x " need $a > 0$ in addition to $\Delta < 0$?

What you expect: Yes. The notes' example never has to say so because a is given. This is the version the exam likes. Ask it right after part (e) of the find- k example.

What he said:

3. *[the one you want answered]*

When the calculator gives the vertex as a decimal, how do we write the vertex form exactly, and does the paper want the fraction?

What you expect: Example 4 gives $V(-2.5, 60.75)$ straight off the calculator. Expect: use $x = -b/(2a)$ as a fraction, exact form on paper, decimals only when the question asks for them.

What he said:

Later. For the HL hour on Wednesday 14 October, Vieta's formulas: the two roots add up to $-b/a$ and multiply to c/a , so you know their sum and product without ever finding them. Worth asking then: do they still hold when $\Delta < 0$ and the roots are complex, and is that another way to see why complex roots come in a conjugate pair? (Yes to both.)

Calculator. Casio: Equation, Polynomial degree 2 for roots. Graph mode, G-Solv (F5): ROOT, MAX or MIN for the vertex, Y-ICPT.

Next. Thursday's double continues 2.2 (examples 4 and 5). No HL hour this Wednesday, fall break. HL returns the week of 12 October with 2.12, sum and product of roots (Vieta): $r_1 + r_2 = -b/a$, $r_1 r_2 = c/a$, then the same idea for any polynomial. Section 2.3, functions, domain and range, also starts that week. Exam II is Thursday 29 October and covers 2.1 to 2.5.